

Natural Capital and the Measurement of Productivity*

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Abstract

Omitting inputs to production, such as natural capital, affects estimates of productivity growth. We show benchmark measurements of productivity growth are strongly correlated with environmental change at the national level, consistent with biased measurement due to the omission of natural capital. We derive theoretical conditions under which the measurement and inclusion of previously-omitted environmental inputs reduces bias and error when estimating productivity growth. Calibrated simulations using data from the World Bank and Penn World Tables suggest including non-renewable natural capital in the measurement of TFP improves measurement for 96 of the 100 countries in our sample.

Keywords: Total Factor Productivity; Natural Capital; Economic Measurement

JEL Codes: E01, O44, O47, Q56

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Innovation and technological change drove an unprecedented increase in economic activity over the past two centuries. At the same time, industrialization muddled societal awareness of fundamental connections between humans and nature. Society’s interest in tabulating the stocks of nature and their associated flows waned, and attention shifted to tracking produced and human capital. Theodore Roosevelt noted this phenomenon at the outset of the 20th century:

“With the rise of peoples from savagery to civilization, and with the consequent growth in the extent and variety of the needs of the average man, there comes a steadily increasing growth of the amount demanded by this average man from the actual resources of the country. And yet, rather curiously, at the same time that there comes that increase in what the average man demands from the resources, he is apt to grow to lose the sense of his dependence upon nature.” – [Roosevelt](#)

Formal systems of economic measurement followed this trend; extant structures provide little guidance for separating nature’s contribution to production from the contributions of other factors. Starting with Simon Kuznets’ report to Congress in 1934, the broader development of statistical, economic, and environmental policy, has focused on measuring output and productivity growth by tracking capital and labor over time ([Kuznets](#)). While governments historically emphasized tracking natural resources for posterity and avoiding overuse, features of the environment beyond land were omitted from the transition to the double entry-based national accounting systems used today. For example, sailors, vessels, and sonar are essential inputs to industrial fishing and are measured by national accountants on a regular basis; equally important is the fish stock and its ability to replenish itself over time ([Hannesson](#)). However, the fish stock as an input to production is often excluded from economic measurement. This continues despite managed fish stocks (within exclusive economic zones) being a textbook example of a non-produced natural asset scheduled for measurement under modern standards like the System of National Accounts ([European Commission and Development](#)).

Here, we examine how the omission of the environment as a factor in production affects a leading metric of economic progress produced under current national accounting standards, total factor productivity (TFP). TFP is defined as the growth in output unexplained by growth in inputs, and it has been a benchmark metric for economy-wide technological progress and innovation for nearly 70 years ([Solow](#)). Changes in TFP over time play a crucial role in shaping governments’ budget projections and policy agendas over long time horizons ([Congressional Budget Office](#)). If natural capital affects production, then the omission of environmental assets from national accounting frameworks induces error when estimating aggregate productivity growth. Failing to account for how changes in the environment determine output may lead governments to overestimate the ability for society to grow without the use of nature or to misattribute the effects to economic policy ([Nordhaus and Tobin](#); [Brandt, Schreyer, and Zipperer](#)).

The first half of this paper demonstrates how omitting inputs from the production function affects residual-based estimates of TFP and discusses why natural capital is often omitted in practice. We provide predictions for the correlation of environmental change with TFP growth when

natural capital is incorrectly omitted from the set of measured inputs used to construct TFP. Next, we test whether a benchmark series for TFP growth is orthogonal to proxies for natural services in a balanced panel of data for 117 countries between 1996 and 2019. We find measured growth is strongly correlated with changes in the modest set of natural capital stocks we consider. The correlations between productivity growth and natural capital remain present when we use alternative productivity measurements from the economics literature or those provided by national statistical agencies.

The second half of the paper characterizes when including natural capital aids measurement of TFP in theory, and then considers whether it does so in practice. We derive conditions under which additional measurement unambiguously reduces bias in estimates of TFP growth. These conditions prove to be restrictive and unlikely to hold in most settings. As the theoretical conditions are not formally testable, we use simulation-based evidence to gauge whether including natural capital is likely to improve measurement empirically. Simulation results suggest that including natural capital produces gains in measurement accuracy. Across simulations calibrated to data from 100 countries, the median level of improvement in measurement, measured as the reduction in root mean squared error for estimated average TFP growth, is 6.6 basis points, or a little over 14 percent of the average TFP growth across all countries during the sample period. For the case of the United States, compounding an error reduction of this magnitude over a standard 10-year budget horizon implies a forecast error for cumulative revenues of a little over \$300 billion in 2025. This figure is two to three times larger than the annual year-on-year forecast error in deficit projections by the U.S. Congressional Budget Office (CBO), or roughly a quarter of the aggregate cost of compliance with the Clean Air Act between 1970 and 1990 ([Congressional Budget Office](#); [U.S. Environmental Protection Agency](#)).

Related Literature: Tobin and Nordhaus motivated the inclusion of natural capital when accounting for growth by appealing to its amenity value (towards their broader measure of economic welfare) as well as the productive value of land ([Nordhaus and Tobin](#)). Subsequent research incorporating the environment in the development accounting literature has primarily focused on improving measures of the share of national income attributable to produced capital (e.g., [Mankiw, Romer, and Weil](#); [Caselli](#); [Caselli and Feyrer](#); [Monge-Naranjo, Sánchez, and Santaaulalia-Llopis](#)). These studies apportion capital income between returns on physical investment and resource rents in order to correct for the effects of the latter when comparing the marginal product of produced capital and its share of national income across countries ([Caselli and Feyrer](#); [Monge-Naranjo, Sánchez, and Santaaulalia-Llopis](#)). Broadly, the literature has highlighted the importance of correcting for resource rents when calculating the returns to capital across countries while noting that the amount of residual output unexplained by factor inputs remains substantial ([Hsieh and Klenow](#)). We add to this literature by characterizing how introducing previously-omitted factors of production has ambiguous effects on the bias of measured output elasticities and factor shares.

A more nascent strain of literature focuses on how including natural capital affects growth accounting-based estimates of productivity. Our contribution is closest to those by [Brandt, Schreyer,](#)

and Zipperer and Manning and Barbier, who study the effects of adjusting existing TFP estimates at the national level to account for the extraction of subsurface minerals and fossil fuels. The authors demonstrate that the effects of these adjustments to measured productivity growth depend on the rate of growth in the use of natural capital resources included on the margin relative to those of other factor inputs. Bastien-Olvera and Moore perform a similar exercise when calibrating a regional integrated assessment model used for adjusting the social cost of carbon to account for non-market ecosystem services. Finally, Freeman, Inklaar, and Diewert highlight the importance of accounting for the fact that not all countries extract the same set of non-renewable resources domestically when comparing productivity levels across countries. This is distinct from our exercise comparing productivity growth within countries across time, but presents a methodology that would allow our calibration exercise to be applied to more granular constructions of TFP (e.g., versus those in the Penn World Tables). We add to this literature by explicitly characterizing the conditions when measurement of non-renewable resources reduces errors in estimated TFP, as opposed to calculating the magnitude of the adjustments induced by inclusion. Our approach can also be generalized to consider the effects of including other potentially omitted capital stocks, such as renewable natural capital.

The remainder of the paper proceeds in three steps. Section 1 tests whether existing TFP metrics are orthogonal to proxies for environmental inputs. We show changes in a modest subset of natural capital stocks are strongly correlated with benchmark series for productivity taken from the academic literature and national statistical agencies, consistent with the notion that relevant environmental inputs are omitted. Section 2 derives sharp conditions under which including a partial accounting of natural capital improves TFP measurement. Section 3 uses data on natural resources from the World Bank’s Changing Wealth of Nations project to simulate how including non-renewable resources, when measuring TFP, affects in-sample error. Section 4 concludes.

1 Residual-Based Productivity Measurement

Neoclassical economics attributes output to two sources: factors of production (factor inputs) and technology. Final output $Y(t)$ is determined by the level of factor inputs, the vector $\mathbf{X}(t)$, and a summary statistic for the level of technology available to society, total factor productivity, $TFP(t)$:

$$Y(t) = TFP(t) \times F(\mathbf{X}(t)). \quad (1)$$

Total factor productivity (TFP) is a theoretical modeling device. It is an index number for how technology determines the production possibility frontier at a given point in time.¹ It is a parsimonious treatment for how forces such as innovation and creative destruction shift what society can produce given available resources (Romer; Aghion and Howitt). Under standard assumptions for

¹We assume throughout this paper that technical change is Hicks neutral (i.e., follows 1). A more general formulation of technical change from a modeling standpoint is the partial derivative of the function determining aggregate output in time.

the production function (e.g., the Cobb-Douglas or transcendental logarithmic forms), growth in output is given by

$$g_Y(t) = g_{TFP}(t) + \sum_{x \in \mathbf{X}(t)} \gamma_x(t) g_x(t), \quad (2)$$

where $g_x(t)$ terms denote growth rates and $\gamma_x(t)$ are the output elasticities for each input x at time t .² The growth accounting equation in (2) states that all changes in output are attributable to factors of production or growth in TFP (Barro). Rearranging the growth accounting equation gives

$$g_{TFP}(t) \triangleq g_Y(t) - \sum_{x \in \mathbf{X}(t)} \gamma_x g_x(t); \quad (3)$$

TFP growth is the output growth unexplained by growth in inputs (Comin). The accounting identity in (2) requires that these two concepts of productivity growth—the growth rate of an index number for technology and the residual portion of growth unexplained by factor inputs—coincide.

In practice, TFP growth is measured using the portion of output growth not explained by growth in a set of *observed* factor inputs, which we group together as the set $\mathbf{I}$. Factor inputs (e.g., produced capital and labor) and final output are measured by national statistical agencies at regular frequencies, allowing for statisticians to take versions of (3) to the data. Residual changes in observed output—those unexplained by growth in observed inputs—are called the Solow residuals (Solow) and are imputed to productivity growth by solving

$$g_A(t) \triangleq \hat{g}_Y(t) - \sum_{i \in \mathbf{I}} \hat{\gamma}_i(t) \hat{g}_i(t), \quad (4)$$

where $\hat{g}_i$ and $\hat{\gamma}_i$ are estimates of growth and output elasticities, respectively. A Solow residual is an estimate of TFP growth. Versions of equation (4) are used by statistical agencies like the U.S. Bureau of Labor Statistics (Meyer and Harper) and the European Statistical Office (European Commission and Eurostat) to construct measures of productivity growth in national accounts, and in academic applications comparing productivity across countries and over time (Feenstra, Inklaar, and Timmer). Statistical agencies often label their versions of the estimand in (4) *multifactor productivity* (as opposed to TFP) to acknowledge that they measure productivity growth beyond the factors they include (Organisation for Economic Co-operation and Development).³

If national accountants were able to calculate the dynamics of all factor inputs and their

²In Section 1 we refer to the $\gamma_x(t)$ terms as output elasticities. As shown in Appendix Section 1, all analysis holds if the $\gamma_x(t)$ terms are reinterpreted as the shares of factors of production in GDP when the production function $F(\mathbf{X})$ exhibits constant returns to scale and factor markets are perfectly competitive (Caselli and Feyrer).

³The OECD manual on productivity measurement actually states this explicitly when defining multifactor productivity: “... [t]his manual uses the MFP [multifactor productivity] acronym to signal a certain modesty with respect to the capacity of capturing all factors’ contribution to output growth” (Organisation for Economic Co-operation and Development).

associated output elasticities, then a natural candidate for $\mathbf{I}$ would be to include all observed stocks to ensure $\mathbf{X} \subseteq \mathbf{I}$. A goal then may be to measure all natural capital stocks and include them all at once (Nordhaus and Kokkelenberg). In the interim, the set of measured inputs $\mathbf{I}$ available to practitioners remains limited. Residual-based approaches to productivity measurement deployed by national statistical agencies and researchers typically do not include environmental inputs (Meyer and Harper; Feenstra, Inklaar, and Timmer; European Commission and Eurostat; European Commission and Development).⁴ These omissions are not active decisions to ignore nature; instead, they reflect limited state capacity and the difficulty in measuring natural capital versus other factors of production. Measuring changes in the flows of services from natural capital (c.f., the number of cycles from a turbine in the case of produced capital or total hours worked in the case of labor) that enter the production function each year is difficult; rates of extraction are often not summary statistics for the flow of services the stocks create (e.g., the fishery example in the introduction). Measurement is further complicated by the difficulty of attributing factor income generated by most forms of natural capital. Many services provided by the environment are public goods (e.g., climate and atmosphere) where the private cost of use is zero, rendering income-based approaches to measuring factor shares infeasible.

To illustrate the effects of omitting natural capital inputs when estimating TFP, consider an economy with four factor inputs: produced capital, labor, non-renewable natural capital, and renewable natural capital, such that $\mathbf{X}(t) = \{K(t), L(t), N_1(t), N_2(t)\}$. Suppose a national accountant includes only two inputs, produced capital and labor, when estimating TFP for this economy. Conditional on her choice of inputs ($\mathbf{I}$) and her estimates of factor input growth rates and output elasticities ($\hat{\mathbf{g}}, \hat{\gamma}$), the Solow residual can be decomposed as

$$g_A(t)|\mathbf{I}, \hat{\gamma}, \hat{\mathbf{g}} = \underbrace{g_{TFP}(t)}_{\text{True innovation}} + \underbrace{\sum_{x \in \{N_1, N_2\}} \gamma_x(t) g_x(t)}_{\text{Omitted natural capital inputs}} + \underbrace{\sum_{i \in K, L} [\gamma_i(t) g_i(t) - \hat{\gamma}_i(t) \hat{g}_i(t)]}_{\text{Measurement error}}. \quad (5)$$

When natural capital inputs are omitted and $\mathbf{X} \setminus \mathbf{I} \neq \emptyset$, (3) is a biased estimator for TFP growth; the contributions of the omitted natural capital inputs are misattributed to TFP because they are spuriously absorbed by the Solow residual.⁵ Correlation between the Solow residual in (5) with changes in a given natural capital stock (or other potentially-omitted input) over time is evidence that the stock in question may be an omitted input in production. Conversely, a precise null of zero correlation between a natural capital stock and estimated TFP growth suggests that the empirical growth accounting equation contains the true set of factor inputs (i.e., $\mathbf{X} \subseteq \mathbf{I}$), and omitting the candidate stock for inclusion is unlikely to bias measures of TFP growth.

⁴At best, some TFP growth estimates adjust for payments to environmental factors that produce marketed goods and services where factor income is observed in government accounts. A unified framework would have to account for non-market services, such as the absorption of man-made pollution, when accounting for the totality of inputs into producing all goods and services humanity consumes.

⁵It is possible measurement error exactly cancels out the omitted terms and the estimator is unbiased even when inputs are omitted. We view this case as unlikely.

1.1 Testing for Omitted Natural Capital Inputs

The decomposition in (5) shows that there are three potential channels that may cause the residual-based TFP estimator to be correlated with natural capital: (1) correlation between natural capital use and unobservable growth in TFP, (2) correlation with omitted natural capital stocks that affect output (i.e., are elements of $\mathbf{X}$), and (3) correlation with measurement error when calculating growth rates or output elasticities of included factors. While an econometrician cannot distinguish among these three potential sources of correlation, she can test whether the Solow residuals are uncorrelated with changes in natural capital stocks. If growth in natural capital is correlated with measured TFP growth, then we cannot rule out channel (2). Finding this form of empirical correlation in the data suggests that natural capital is more likely to play a role in output.

In a world with perfect information over the true growth rates of factor inputs and TFP, we could test whether a given natural capital stock is an omitted variable (channel 2) in isolation. However, as we only observe g_A jointly as the sum of the three terms in (5), we are restricted to testing whether the correlations between growth in natural capital stocks and *measured* TFP are nonzero. This leads to two potential sources of type one errors where we correctly reject the null hypothesis of no correlation between g_A and g_{N_i} but spuriously attribute it to N_i being an omitted input in production. These cases correspond to channels (1) and (3); when g_{N_i} is systemically correlated with true innovation or measurement error in the growth accounting equation. This also brings about the potential for type two errors: when g_{N_i} is an omitted variable in the growth accounting equation and correlated with measurement error or TFP growth. It is possible these correlations are offsetting. The econometrician may then correctly fail to reject the null hypothesis of no correlation between g_A and g_{N_i} , but spuriously attribute this to natural capital playing no role in production.⁶

1.2 Estimating Equation

To test for correlations, we regress measured national TFP growth ($g_{A_{j,t}}$) from the Penn World Tables (PWT, [Feenstra, Inklaar, and Timmer](#)) on growth in the seven country-level natural capital stocks ($g_{N_{n,j,t}}$) measured by the 2024 version of the World Bank’s The Changing Wealth of Nations (CWON) dataset. Equation (6) shows the empirical analog of (5) that we take to the available data. We estimate (6) on a balanced panel of 117 countries over the 1996-2019 period:⁷

⁶Several features of measurement may bring about these spurious results in practice. First, estimating output elasticities is nontrivial even for cases where factor income can be reasonably attributed ([Caselli and Feyrer](#)). Second, the lack of attributed factor income makes it difficult to delineate which changes in measured natural capital stocks constitute proxies for changes in flows entering production. Finally, while measures of some forms of innovation exist (e.g., patent filings and seed varieties, used as proxies for innovation in [Acemoglu et al.](#) and [Moscona and Sastry](#) respectively), underlying correlations between environmental change and TFP growth are not falsifiable.

⁷The CWON dataset in its raw form contains an unbalanced panel of natural capital stocks in each country covering 1995-2020. We impute missing values as zero changes from the last period where measurements were available leaving us with a balanced panel of growth in natural capital stocks between 1996 and 2020 (see Appendix Section 2.2 and 2.3 for a discussion of the imputation process for missing data along with robustness checks to ensure the results are not an artifact of this process). Data on total factor productivity levels in the PWT version 10.01 only cover years out to 2019, as such our estimating sample covers 1996-2019.

$$g_{A_{j,t}} = \sum_{i \in \mathbf{I}} \beta_i g_{N_{i,j,t}} + \sum_{z \in \mathbf{Z}} \theta_z g_{z,j,t} + \delta_j + \zeta_t + \varepsilon_{j,t} \quad (6)$$

where j and t index countries and years respectively. The set $\mathbf{I}$ covers the seven natural capital stocks measured at the country level present in the CWON, all measured in physical quantity units: urban land, agricultural land, timber land, non-timber forests, mangroves, fishery biomass, and hydroelectric power generation.⁸ Coefficients β_i govern their relationship with TFP growth. The summation over terms $g_{z,j,t}$ allows for specifications that control for growth in other factor inputs such as produced capital, employment, labor's share of income, and human capital taken directly from the PWT (Feenstra, Inklaar, and Timmer). Terms (δ_j, ζ_t) allow for country and year fixed effects (labeled TWFE in Table 1), and $\varepsilon_{j,t}$ are error terms capturing residual idiosyncratic variation in TFP growth.

The β_i terms are the estimands of interest. While (6) resembles (5), the β_i coefficients on natural capital growth rates are not the associated output elasticities. Inference on these coefficients does not test for causal effects of changes in natural capital stocks on productivity growth. We do not claim that they separately identify the omitted variable bias we seek to test from other potential channels for correlation. Rather, these coefficients identify the (conditional) covariance between growth in natural capital stocks and growth in measured TFP. Failing to reject the joint null hypothesis of interest:

$$H_0 : \beta = \mathbf{0} \quad (7)$$

only indicates that we cannot rule out that the set of natural capital stocks we consider is uncorrelated with the combination of components in the decomposition of TFP growth in (5).

Three assumptions must hold in order to interpret the rejection of the null hypothesis in (7) as evidence that natural capital inputs are omitted when calculating TFP. First, measurement error in the (weighted) factor input growth terms used to construct estimates of TFP is uncorrelated with growth in natural capital stocks. Second, changes in natural capital stocks are a good proxy for changes in the flow of services they produce. Third, the rate of true technical change g_{TFP} is conditionally uncorrelated with the changes in environmental inputs to production that we consider. These assumptions are strong. However, their failure would not indicate that natural capital is irrelevant for national accountants. Rather, if the first assumption fails, then better measurement of natural capital improves the measurement of factor inputs that currently enter national accounts even if measurement of natural capital provides no direct value when calculating productivity growth. If the second assumption fails, then it is a powerful argument for adjusting the natural capital stocks we measure in order to keep better track of the aspects of the environment that are important for the economy. Finally, if the third assumption fails, then measurement of natural

⁸A full description of how the data on individual natural capital stocks are constructed in the CWON and how we clean these data prior to estimation along with the list of countries covered in our analysis can be found in Appendix 2.2.

capital is a first-order priority for national accountants in order to determine how nations may adjust the use of the environment to accelerate true productivity growth.

1.3 Reduced-Form Results

Table 1 displays estimates of $\hat{\beta}$ from various specifications of (6), each of which shows how a change in the growth of a given natural capital stock affects expected growth in TFP. Each column gives two tests of the null hypothesis in (7) that the vector of coefficients on growth in natural capital $\beta = [\beta_1, \dots, \beta_7]'$ is the zero vector. Failing to reject H_0 suggests that the collection of renewable natural capital stocks in CWON plays no role in the production function, and omitting them is unlikely to bias current metrics of TFP.

The bottom rows of table 1 show p -values from two Wald tests of this joint null hypothesis across all seven natural capital stocks we consider. The first row constructs test statistics using asymptotic standard errors (clustered at the country level), while the second uses a semi-parametric wild bootstrap approach (Wu). We can reject that the natural capital stocks we consider are uncorrelated with productivity growth at traditional significance levels under the tests using asymptotic standard errors across all specifications, and the tests using the wild bootstrap approach are always at least statistically significant at the 10 percent level. Furthermore, almost all point estimates are positive, the expected sign if the natural capital stocks we consider are beneficial inputs to production and covary positively with output.⁹ While the estimated joint effects of changes in natural capital appear precise, the overall residual variation explained by adding natural capital on the margin is limited. Including natural capital increases model R^2 by 0.006 (3 percent over baseline levels) for the benchmark fixed effects models in columns (3) and (4).

Robustness: Partial correlations and individual tests of coefficient precision suggest services from land are the environmental inputs most likely to be material for current TFP metrics, consistent with the suggestions for improving national accounting given in the latest edition of the PWT (Feenstra, Inklaar, and Timmer). To ensure missing land rents are not the sole drivers of the results here, we re-estimate equation (6) after excluding urban, agricultural, and timber land from the set of natural capital stocks. Under this specification focusing on a subsample of four renewable natural capital stocks outside of land, our estimates lose precision and we are only able to reject the null hypothesis at the 10 percent level in about half of the specifications we consider (see Appendix 2.3). This confirms the suggestions in Feenstra, Inklaar, and Timmer that better accounting for land rents in national accounts is likely a fruitful direction for expansion. Appendix 2.3.2 provides several additional checks of the robustness of our results to alternative choices for the set of natural capital stocks considered (e.g., excluding natural capital stocks with a high share of missing or imputed values, expanding them to include non-renewable natural capital, excluding

⁹The negative coefficient on forest cover may be due to increases coinciding with declines in logging or forestry activities which are counted in final output. This does not per se imply that forests are a drag on the economy, but is rather indicative that the services they produce (outside of timber) are not captured by measured output. Generally, any societal benefit produced by a given natural capital stock that is conditionally uncorrelated with market output will not cause a correlation with TFP growth in this exercise.

only urban land, and considering alternative sample periods). Appendix 2.3.3 shows our results hold for alternative choices of outcome variables capturing productivity growth. We can reject the joint null when (6) is re-estimated on a productivity growth panel for the 27 European Union nations plus Norway measured by EUROSTAT and when we use measured growth in output as the outcome variable and control for growth in factor inputs such as produced capital, employment, and human capital.

2 The Marginal Inclusion of Potentially Omitted Inputs

Section 1 provides suggestive evidence that natural capital is an omitted input in existing measures of TFP. Now, we turn to examining when including changes in previously-omitted inputs in the growth accounting equation improves residual-based estimates of productivity growth. Intuition suggests that we may expect that including an omitted determinant of output in the growth accounting equation will improve estimates of TFP. However, Appendix 1.4 shows that including an omitted input is only guaranteed to reduce bias under very strong assumptions. To build intuition, consider when the residual-based estimator is applied to our four-input example economy. Suppose a national accountant is able to measure growth in output, labor, produced capital, and non-renewables as well as the associated output elasticities (or factor shares) without error, but cannot measure growth in renewable resources over time (g_{N_2}). If the accountant elects to omit non-renewables (g_{N_1}) when calculating TFP growth, substituting this into (4) yields the estimand

$$\hat{g}_A(t) \triangleq g_{TFP}(t) + \gamma_1 g_{N_1}(t) + \gamma_2 g_{N_2}(t) \quad (8)$$

for TFP growth. If the accountant instead includes non-renewable in the residual-based estimator, the estimand for TFP is

$$\tilde{g}_A(t) \triangleq g_{TFP}(t) + \gamma_2 g_{N_2}(t) \quad (9)$$

It is ambiguous whether (9) is a less biased estimate of g_{TFP} than (8) without making assumptions on the sign of growth in renewable inputs over time. Appendix 1.4 generalizes this result to the case where multiple inputs may be simultaneously omitted; the bias in the residual-based estimator of average TFP growth is reduced only when all omitted terms share a sign in expectation. Criteria for improving in-sample metrics of goodness of fit (e.g., root mean squared error) are even more stringent. They depend on higher moments of growth in the omitted stocks and the set of included inputs. Including natural capital on the margin, even with perfect measurement of growth and unbiased and consistent estimations of output elasticities, are only guaranteed to improve estimates of TFP when it leads the estimator to be correctly specified (i.e., $\mathbf{X} \subseteq \mathbf{I}$) or the pairwise growth rates of omitted natural capital inputs share very similar covariance structures.¹⁰

The result that the marginal inclusion of natural capital need not reduce bias in estimates

¹⁰See Proposition 2 in Appendix 1.4

of TFP growth is new to the growth accounting literature. For example, existing work by [Brandt, Schreyer, and Zipperer](#) and [Manning and Barbier](#) examines how partial inclusion of natural capital inputs affects income-based growth accounting estimates of TFP against baseline measures. However, because of the results shown here, it is not possible to know whether these adjustments for non-renewable resources bring the resulting alternative estimates closer to the rate of true productivity growth over time.

3 Simulating the Effects of Incorporating Natural Capital when Estimating TFP

Section 2 examined the theoretical effects of reducing the number of omitted factor inputs on the performance of the Solow residual as an estimator of TFP. This section evaluates this question using the available data on growth in factor inputs, natural capital stocks, and output. We take a version of our running four-input economy to the data and examine whether including non-renewables on the margin improves residual-based estimates of TFP.

3.1 Setup

The goal of our exercise is to calibrate a version of the four-factor economy from Section 2 at the country level for the panel of countries in Section 1. We consider a case where a national accountant seeks to measure TFP growth for this economy, but measures only growth in the usage of produced capital, labor, and non-renewable capital, N_1 . We assume she suspects that services from renewable capital (N_2) are important inputs in production, but that she is unable to measure how their use has changed over time. Knowing this, she must choose whether to include growth in the use of non-renewable resources, g_{N_1} , when taking Equation (4) to the data to estimate TFP growth. Her criteria for including non-renewable resources in this thought experiment is whether it lowers expected in-sample mean squared error when estimating average TFP growth in her data, which we assume covers 25 years of growth in output and factor inputs.¹¹

Data Generating Process for Output: Output growth in each country comprises growth in TFP, labor, produced capital, and two forms of natural capital inputs (non-renewables N_1 and renewables N_2). Joint observations of growth rates for all determinants of output growth are assumed to be independent and identically distributed draws of vectors $\mathbf{g}_{j,t}$ from a fixed distribution for each country with mean $\boldsymbol{\mu}_j$ and variance Σ_j . Output growth in each simulated country j is generated by

$$g_{Y,j,t} = \mathbf{g}_{j,t}' \boldsymbol{\gamma}_j \quad (10)$$

where $\boldsymbol{\gamma}_j \triangleq [1, \gamma_K, \gamma_L, \gamma_1, \gamma_2]'$ is the vector of output elasticities for country j .

¹¹We round up to 25 from the size of 24 years in our estimating sample.

Calibrating Country-Level Growth: We assume at the country level that growth in TFP (g_{TFP}) is uncorrelated with growth in other factors. We calibrate the mean (μ_j) and variance (Σ_j) governing the joint distribution of growth terms $g_{j,t}$ in each country based on sample moments listed in Table 2. For natural capital, we follow the World Bank CWON methodology (World Bank) and construct aggregate Törnqvist quantity indices for non-renewable and renewable resource use at the country level over time (see Appendix Section 3). We set the parameters governing average growth in the use of natural capital equal to the sample means of growth in these indices between 1996 and 2019 (Table 2 rows one and two). Expected growth in produced capital services, labor, and TFP are set equal to their sample analogs in the Penn World Tables (Feenstra, Inklaar, and Timmer) over the same 1996-2019 period. Higher order moments in Σ are set based on the empirical covariances.

Output Elasticities: We set the output elasticity of services from renewable natural capital, $\gamma_2 = 0.05$ based on the analysis in Appendix Section 3.6. In turn, to induce an overall output elasticity of 0.1 for natural capital (following Arrow et al.), we set the output elasticity of services from non-renewable natural capital, γ_1 , to 0.05. Sensitivity analyses of our results with regard to the output elasticities for natural capital are in Appendix 4.5.1.

Procedure for Estimating TFP Growth: The national accountant observes 25 years' worth of data for the subset, K, L , and N_1 , of growth in factor inputs along with output growth in a given country. We assume she does not observe the distribution of national income earned across factors, and therefore cannot estimate output elasticities using factor shares. This ensures our thought experiment is consistent with estimating the Solow residual in empirical settings, where factor income generated by natural capital such as common pool resources is not attributed to the environment in national accounts and is likely unobservable to an econometrician (Manning, Taylor, and Wilen).

To construct a measure of TFP growth which does not rely on growth in non-renewable resources, the national accountant estimates:

$$g_Y = \hat{\gamma}_A + \hat{\gamma}_K g_K + \hat{\gamma}_L g_L + \epsilon \quad (11)$$

on time series of data for a given country using ordinary least squares (OLS). Her estimator for average TFP growth based only on growth in produced capital and labor is the intercept term $\hat{\gamma}_A$. We refer to this estimator as model 1. She then estimates the analog for the case including non-renewable resources:

$$g_Y = \tilde{\gamma}_A + \tilde{\gamma}_K g_K + \tilde{\gamma}_1 g_{N_1} + \tilde{\gamma}_L g_L + \nu \quad (12)$$

on time series for growth in $\{K, L, N_1\}$. The OLS-based estimator for average TFP growth inclusive of non-renewable capital is $\tilde{\gamma}_A$. We refer to this estimator as model 2.

We then calculate the expected RMSE for $\hat{\gamma}_A$ and $\tilde{\gamma}_A$ in each country by evaluating models 1 and 2 at the sample analogs of μ and Σ and comparing these estimates relative to the calibrated

values, $\mathbb{E}[g_{TFP}]$:

$$\Delta\text{RMSE}|\boldsymbol{\mu}, \Sigma \triangleq \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} - \sqrt{\mathbb{E}[(\hat{\gamma}_A - \mathbb{E}[g_{TFP}])^2|\boldsymbol{\mu}, \Sigma]} \quad (13)$$

Applying this process to each country in the sample with sufficient data yields a cross section of 100 country-level observations of ΔRMSE .¹²

3.2 Results

For 96 out of the 100 countries for which we have sufficient data, including non-renewable resources lowers the RMSE for estimated TFP growth (Figure 1). A negative ΔRMSE (green bars to the left of $x = 0$) indicates that including non-renewable resources improves the accuracy of a national accountant's estimate of TFP using the residual-based method under the mean squared error criterion. A positive ΔRMSE indicates including non-renewables increases the RMSE of estimated TFP growth and worsens measurement (blue bars). The median and mean improvements in the accuracy measure are sizable: reductions of 6.6 and 32 basis points (one hundredth of a percentage point, abbreviated as bps) respectively. Improvements are disperse (standard deviation of 65 bps) and left-skewed. The countries with the greatest improvements in terms of how much including non-renewables reduces RMSE when estimating TFP growth are Kenya, Laos, Israel, Tanzania, and Niger. The magnitude of the median improvement is a little over 14 percent of average TFP growth across all countries over the 1996-2019 period. For the four countries where including non-renewables raises RMSE, the deterioration in fit is small (less than 1 basis point both on average and at the median). The greatest RMSE increase occurs for Chile, where RMSE rises by 3.3 bps.¹³

To decompose the contribution of individual components of output growth to changes in measurement accuracy, we regress ΔRMSE on the individual components of $\boldsymbol{\mu}_j$ and Σ_j . This gives a reduced-form sense of the elements of input growth most important in determining the magnitude of accuracy improvements on average.¹⁴ Appendix Table 18 shows that ΔRMSE is increasing in the country-level growth rates of renewable and non-renewable resource use, and decreasing in the correlations between growth in non-renewable resource use and those for labor and produced capital.¹⁵ Thus if the world looks increasingly like Spain or Kyrgyzstan—nations where growth in the use of non-renewables is negatively correlated with produced capital and labor, then including non-renewable resources in natural accounts is more likely to improve estimates of TFP growth. In contrast, if the world becomes more like Chile or Indonesia, where growth in the use of non-

¹²See Section 4.1 of the Appendix for the full procedure.

¹³Chile is an empirical case where the two components of bias in (11), stemming from omitting two variables, happen to offset each other almost perfectly, causing the bias when only renewables are omitted in (12) to be larger in magnitude.

¹⁴The true gradients for each country can be calculated using the closed-form solutions for ΔRMSE we derive in Sections 4.2 and 4.3 of the Appendix.

¹⁵Three of these four determinants of ΔRMSE are significant at traditional (5 percent) levels when ΔRMSE is regressed on the elements of $\boldsymbol{\mu}_j$ and Σ_j that enter the RMSE calculations. The joint test is significant at the 0.1 percent level. See Section 4.4 of the Appendix for details and reduced-form analysis of other how other sample moments determine ΔRMSE .

renewables is positively correlated with produced capital and labor, then adding non-renewables to national accounts will have a limited impact on existing estimates of TFP growth.

Figure 2 shows scatter plots illustrating the cross-sectional variation in the relative performance of the estimators against individual country-level characteristics such as trends in resource use and baseline levels of calibrated productivity growth. Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in the use of non-renewable and renewable resources (g_{N_1} and g_{N_2}) at the country level. Of the 100 countries in our sample, 68 countries exhibit positive average growth in their use of non-renewables, and 47 exhibit positive average growth in the use of renewable inputs. Importantly, panels 2a and 2b show that average growth in non-renewables is about an order of magnitude greater than growth in renewables. This proportionally larger growth in non-renewable resources is responsible for substantially more output growth than that of renewables, even when the contribution of both resources to output in levels is similar. In turn, leaving out non-renewables on the margin leaves much more unexplained output in model 1 relative to model 2 where only renewables are omitted.¹⁶

Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process (set as the observed value in the PWT). Panels 2e and 2f show the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. The median (mean) improvement is 10 (22) percent of baseline error when both natural capital stocks are omitted, and rises to more than 40 percent for over a quarter of the countries in the cross section of hypothetical outcomes. Reduction in the bias of the estimators drives most of the improvement rather than a smaller expected variance. Bias drives 98.5 percent of RMSE on average when non-renewables are excluded (model 1), and 81 percent of bias on average when non-renewables are included (model 2).¹⁷ Additional cross sectional analysis examining how the country-level outcomes in our simulations relate to other factors that play a role in measuring productivity (e.g., statistical capacity and the share of GDP stemming from resource rents) can be found in Appendix 4.4.

The resulting improvement for most countries is of a magnitude that is policy-relevant. The median reduction in RMSE across the 100 countries in our sample is 6.6 basis points. This translates to an annual forecast error of roughly 0.07 percentage points when historical estimates are used to forecast average growth in the future. Applying this median value to the United States, when this forecast error is compounded over ten years, the standard CBO budget horizon, at current levels of output, forecast cumulative revenues would deviate from projections by approximately \$300 billion.¹⁸ This forecast error is roughly one quarter of the aggregate social cost of compliance with the Clean Air Act in the United States between 1970 and 1990 in magnitude, one of the

¹⁶Figure 4 in Appendix 4.4 shows a version of these two panels for all 12 sample moments affecting Δ RMSE at the country level.

¹⁷See Appendix 4 for the derivation and decomposition of the changes in RMSE in terms of changes in bias and variance.

¹⁸See Appendix 5 for breakdown of the back-of-the-envelope calculation here. The deviation of projections for the United States would grow to slightly under \$1.2 trillion if we were to instead apply our U.S.-specific point estimate of a 26 basis point change in RMSE when including non-renewables on the margin .

broadest environmental policies enacted globally to date ([U.S. Environmental Protection Agency](#)).

Comparison to existing results in the literature: The magnitude of changes in estimates of annual TFP growth are similar to those produced by [Brandt, Schreyer, and Zipperer](#), who adjust existing productivity metrics for a subset of non-renewable resources considered here. The authors find including non-renewable capital changes point estimates of annual TFP growth by an average of 2.3 basis points for a panel of 21 countries between 1986 and 2008. [Manning and Barbier](#) find adjusting the growth accounting equation to account for natural capital increases the amount of output growth unexplained by factor inputs by 87 basis points on average for countries with negative unadjusted TFP growth and decreases it by 65 basis points on average for countries where TFP growth is positive. [Freeman, Inklaar, and Diewert](#) use an earlier version of the Changing Wealth of Nations dataset to examine how inclusion of subsurface minerals and fossil fuels affects estimates of TFP relative to that of the United States. They find omitting non-renewable resources can bias TFP levels by up to 50 percent for resource-intensive economies such as those in OPEC or those with large mining sectors, in-line with the proportional reductions in error we show in Figure 2f. Finally, our estimates for the median reduction in RMSE when estimating TFP growth is about one-third the magnitude of the latest revisions of TFP estimates (0.2 percentage point) made by the United States’ Bureau of Labor Statistics for 2024 ([U.S. Bureau of Labor Statistics](#)).

Sensitivity Analysis: We present sensitivity analysis of our simulation results to alternative choices of output elasticities for natural capital and sample periods for moments governing input growth in Appendix 4.5. Appendix 4.5.1 performs a Monte Carlo analysis of our exercise across ten million alternative pairs of output elasticities drawn from an iid uniform distribution on the interval $[0, 0.1]$. The median (mean) RMSE reductions fall to 2 (28) basis points and are negative under about 30 percent of alternative parameterizations, highlighting the importance of better measurement and apportionment of income generated by natural capital to generate better estimates of their contributions to output. Appendix 4.5.2 shows that re-estimating the sample moments in Table 2 based on two partitioned subsamples of the original 1996-2019 data leads to larger estimates of RMSE improvements across the board and minimal qualitative changes. The set of countries for which these improvements are negative remains quite small in both cases (ten and four countries respectively in the early and late periods).

4 Conclusion

Fifty-four years after [Nordhaus and Tobin](#), research continues to highlight new channels by which environmental change affects society and welfare. An increasing share of this research has focused on responding to high-level calls for improved welfare measures, e.g., the U.N. Secretary General’s push for measures “Beyond GDP” ([United Nations Statistics Division](#)). Economists have long recognized the importance of such measures, and there is a rich literature on complements to GDP that would better-account for the determinants of welfare beyond market activity ([Fleurbaey](#); [Jorgenson](#)).¹⁹

¹⁹[Conte, Addicott, and Millerhaller](#) provide micro-founded support for this position with regard to the environment, demonstrating that reductions in natural capital stocks generate real income effects for consumers.

This paper gives evidence that measurement of natural capital benefit governments irrespective of goals beyond GDP. When natural capital affects production, omitting it from national accounts biases estimates of economic growth. This supports the original hypothesis by Nordhaus and Tobin in 1973 that leaving things out (“treating as free things that are not really free”) of national accounts “... gives the wrong signals for the directions of economic growth” in addition to affecting measurement of national welfare (Nordhaus and Tobin). While measuring the environment will remain important for assessing the sustainability of development, regulatory compliance, and public health, we show here systematically connecting changes in the environment to economic statistics will benefit existing measurement that is already used to guide policy.

Table 1: TFP Growth vs. Growth in Renewable Natural Capital Stocks

Outcome: $\Delta \ln(TFP_{j,t})$	(1)	(2)	(3)	(4)	(5)	(6)
Urban Land	0.301 (0.095)	0.303 (0.098)	0.205 (0.067)	0.215 (0.075)	0.149 (0.062)	0.060 (0.118)
Timber Land	0.164 (0.090)	0.166 (0.075)	0.071 (0.074)	0.075 (0.071)	-0.003 (0.068)	0.026 (0.104)
Agricultural Land	0.027 (0.025)	0.043 (0.025)	0.030 (0.022)	0.036 (0.023)	0.041 (0.021)	0.043 (0.017)
Forest Cover	-0.067 (0.094)	-0.111 (0.091)	-0.005 (0.079)	0.012 (0.083)	0.083 (0.053)	-0.027 (0.076)
Mangroves	0.122 (0.244)	0.067 (0.242)	0.357 (0.299)	0.327 (0.322)	0.318 (0.367)	-0.108 (0.427)
Fishery Biomass	0.112 (0.098)	0.026 (0.077)	0.094 (0.108)	0.072 (0.101)	0.047 (0.135)	-0.104 (0.108)
Hydropower	0.011 (0.004)	0.011 (0.004)	0.009 (0.004)	0.009 (0.004)	0.006 (0.003)	0.006 (0.003)
Controls for ...						
Factor Inputs	No	Yes	No	Yes	Yes	Yes
TWFE	No	No	Yes	Yes	Yes	Yes
Idiosyncratic Trends	No	No	No	No	Yes	No
Lags of Outcome	No	No	No	No	No	Yes
Wald	0.007	0.003	0.007	0.005	0.002	0.039
Bootstrap Wald	0.089	0.056	0.044	0.018	—	0.037
# Observations	2,808	2,808	2,808	2,808	2,808	2,457
# Countries	117	117	117	117	117	117

Note: The outcome variable is the change in the logarithm of the $RTFP^{NA}$ series in version 10.01 of the Penn World Tables ([Feenstra, Inklaar, and Timmer](#)). All independent variables enter as log changes. Standard errors clustered at the country level are shown in parentheses. Rows under the “Controls for ...” heading denote whether covariates controlling for factor inputs (capital services, labor, human capital, and the labor share), country and year fixed effects (TWFE), and idiosyncratic time trends at the country level are included. The last column controls for two lags of the dependent variable $\Delta \ln(TFP_{j,t})$ and is estimated using the Arellano-Bond (1991) dynamic panel estimator and reports robust (as opposed to clustered) standard errors. Row “Wald” displays p values from the joint null hypothesis that coefficients on all natural capital growth terms are zero. Standard errors for the Row “Bootstrap Wald” shows p -values from wild bootstrap tests of the joint null hypothesis ([Wu](#)). No results are reported for this test in column (5) due to the idiosyncratic trend terms.

Table 2: Simulation Moments and Sources

Description	Population Moment	Sample Moment	Source
Growth in Non-Renewable Capital Services	$\mathbb{E}[g_{N_1}]$	$\bar{g}^R$	CWON
Growth in Renewable Capital Services	$\mathbb{E}[g_{N_2}]$	$\bar{g}^{NR}$	CWON
Growth in Produced Capital Services	$\mathbb{E}[g_K]$	$\bar{g}_K$	PWT
Growth in Labor Inputs	$\mathbb{E}[g_L]$	$\bar{g}_L$	PWT
Growth in TFP	$\mathbb{E}[g_{TFP}]$	$\bar{g}_{TFP}$	PWT
Covariances of Growth Terms	Σ	$\hat{\Sigma}$	CWON & PWT

Note: Subscripts indexing each object by country are dropped in the first two columns; all population moments are assigned the sample analogs on a country-by-country basis. Bars denote sample averages while hats denote sample analogs for the covariance matrix Σ . CWON refers to the World Bank *Changing Wealth of Nations* database ([World Bank](#)). PWT refers to the Penn World Tables version 10.01 ([Feenstra, Inklaar, and Timmer](#)).

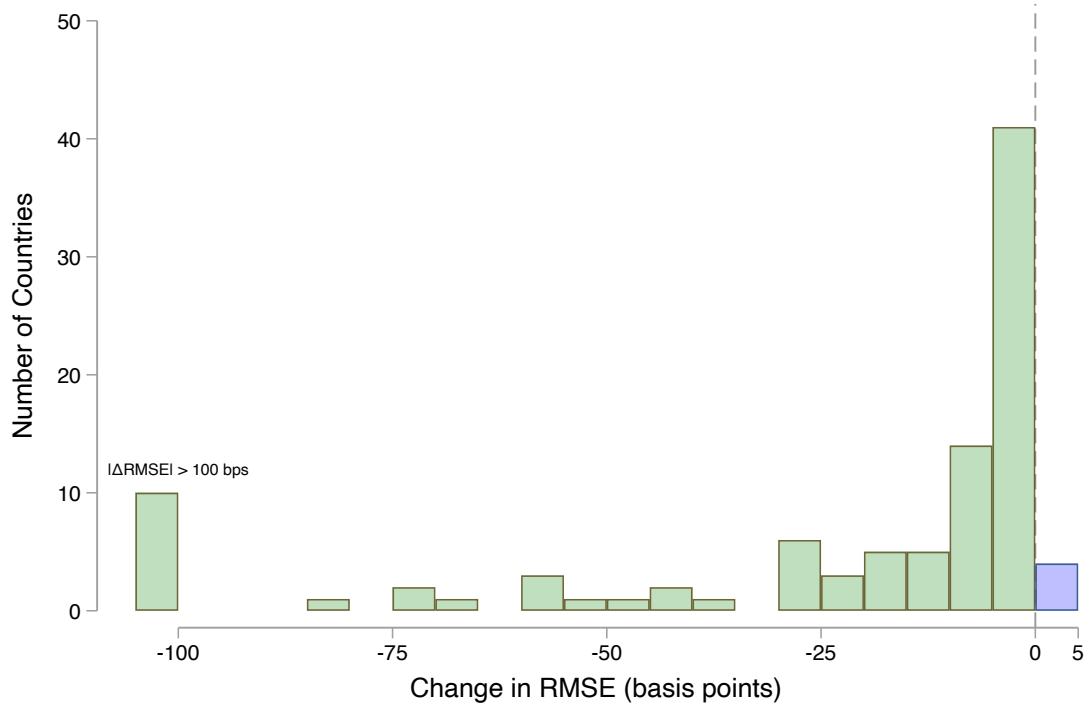


Figure 1: Histogram of the reductions in mean squared error— ΔRMSE —when adding g_{N_1} to the Solow residual-based TFP estimator for each of 100 countries in the sample. The 10 countries with RMSE reductions larger than 100 basis points in magnitude are topcoded and added to the leftmost bin in the histogram. See Appendix Table 15 for a full list of country-level sample moments and changes in RMSE.

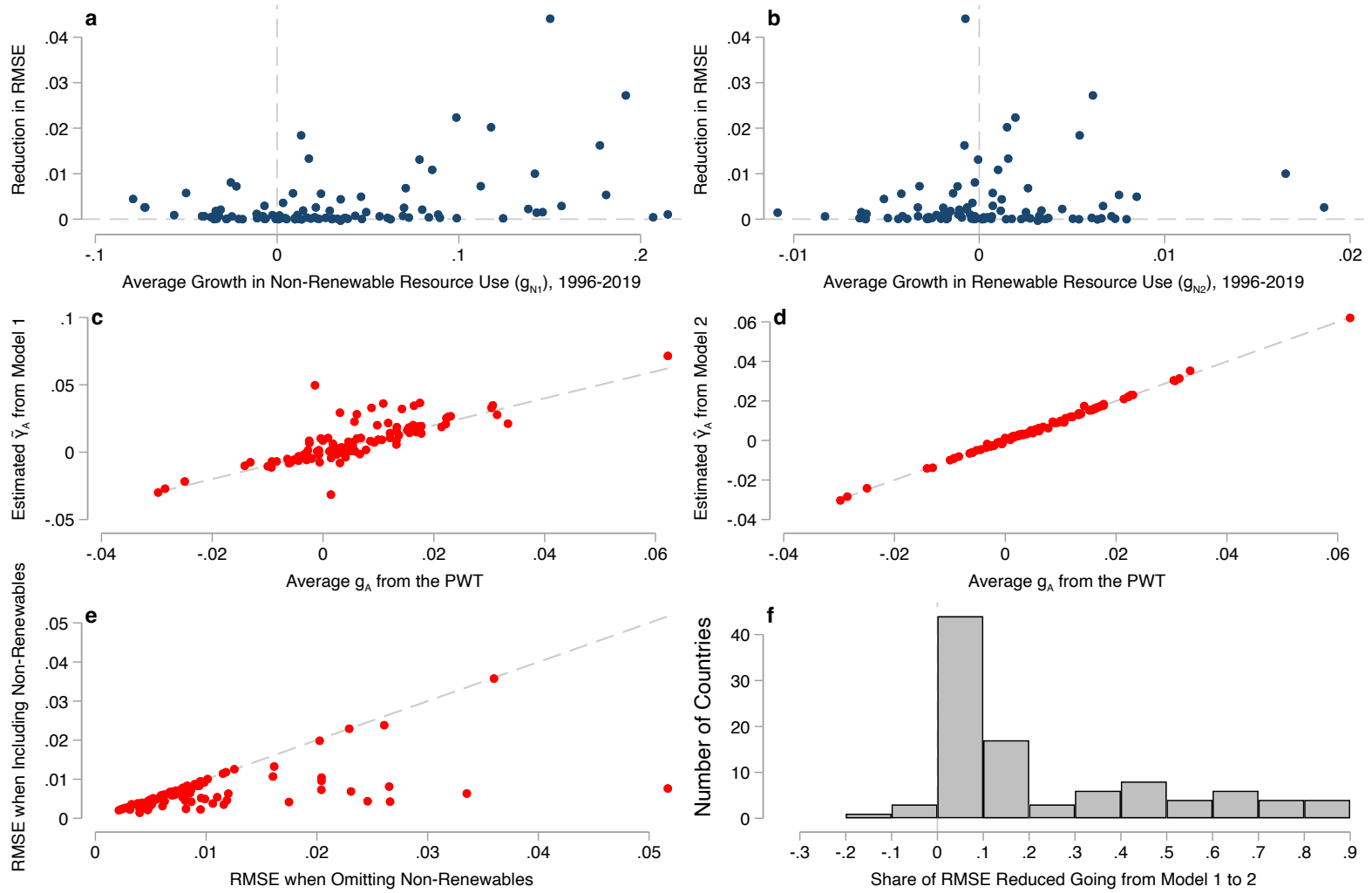


Figure 2: Panels 2a and 2b plot improvements in estimates of TFP growth against average growth in non-renewable and renewable resource use at the country level between 1996 and 2019 for our panel of 100 countries. Panels 2c and 2d show point estimates of TFP growth based on historical moments from the two estimators against the true data generating process. Panels 2e and 2f shows the improvement from including non-renewable resources against the baseline error in the model omitting natural capital as well as the relative size of this improvement. Dashed lines in panels 2a, 2b, and 2f are plotted at $x = 0$ and $y = 0$. Dashed lines in panels 2c-2e show the 45 degree line.

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